

Continuity of Functions at a Point – Questions with Step-by-Step Solutions

1. Check whether $f(x) = x^2 + 3x + 1$ is continuous at $x = 2$.

- Step 1: Find the function value: $f(2) = 2^2 + 3(2) + 1 = 4 + 6 + 1 = 11$.
- Step 2: Compute the limit as $x \rightarrow 2$: $\lim_{x \rightarrow 2} (x^2 + 3x + 1)$. Since it is a polynomial, substitute $x = 2$ to get 11.
- Step 3: Compare: $\lim_{x \rightarrow 2} f(x) = 11 = f(2)$.
- Conclusion: The function is continuous at $x = 2$.

2. Determine whether $f(x) = \sin(x)$ is continuous at $x = \pi$.

- Step 1: Function value: $f(\pi) = \sin(\pi) = 0$.
- Step 2: Limit: $\lim_{x \rightarrow \pi} \sin(x) = \sin(\pi) = 0$.
- Step 3: Compare: $\lim_{x \rightarrow \pi} f(x) = 0 = f(\pi)$.
- Conclusion: The function is continuous at $x = \pi$.

3. Check the continuity of $f(x) = 1/x$ at $x = 1$.

- Step 1: Function value: $f(1) = 1/1 = 1$.
- Step 2: Limit: $\lim_{x \rightarrow 1} 1/x = 1$.
- Step 3: Compare: $\lim_{x \rightarrow 1} f(x) = 1 = f(1)$.
- Conclusion: The function is continuous at $x = 1$.

4. Is the function $f(x) = \sqrt{x}$ continuous at $x = 4$?

- Step 1: Function value: $f(4) = \sqrt{4} = 2$.
- Step 2: Limit: $\lim_{x \rightarrow 4} \sqrt{x} = 2$.
- Step 3: Compare: $\lim_{x \rightarrow 4} f(x) = 2 = f(4)$.
- Conclusion: The function is continuous at $x = 4$.

5. Check whether $f(x) = e^x$ is continuous at $x = 0$.

- Step 1: Function value: $f(0) = e^0 = 1$.
- Step 2: Limit: $\lim_{x \rightarrow 0} e^x = 1$.

- Step 3: Compare: $\lim_{x \rightarrow 0} f(x) = 1 = f(0)$.
- Conclusion: The function is continuous at $x = 0$.